

Rita Torres de Sousa

Rita Torres de Sousa holds a Bachelor's degree in Health Sciences, a Master's degree in Bioinformatics and Computational Biology, and a Ph.D. degree in Informatics from the University of Lisbon. Since her master's thesis, her research has focused on integrating life sciences expertise with computational skills to develop new approaches to learn from complex biomedical data and discover new knowledge. More specifically, her research interests include biomedical knowledge graphs, semantic similarity, machine learning and explainable artificial intelligence.

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Education

PhD in Informatics, Approved with Distinction and Prizes

FACULTY OF SCIENCES OF THE UNIVERSITY OF LISBON

- Thesis: Semantic perspectives for learning over biomedical knowledge graphs.

Lisbon, Portugal

09/2019 - 07/2024

International Semantic Web Research Summer School

STI INTERNATIONAL

Bertinoro, Italy

07/2023

Master in Bioinformatics and Computational Biology (BBC), 19/20

FACULTY OF SCIENCES OF THE UNIVERSITY OF LISBON

- Dissertation: Evolving meaning - using Genetic Programming to learn similarity perspectives for mining biomedical data (19/20).

Lisbon, Portugal

09/2017 - 07/2019

Bachelor in Health Sciences, 17/20

UNIVERSITY OF LISBON

Lisbon, Portugal

09/2014 - 07/2017

Professional Experience

Data and Web Science Group, University of Mannheim

RESEARCHER

Mannheim, Germany

01/2024 - Present

Data and Web Science Group, University of Mannheim

VISITING RESEARCHER (DAAD FUNDING PROGRAM: STIBET-DOKTORANDEN)

Mannheim, Germany

09/2022 - 11/2022

LASIGE, Faculty of Sciences, University of Lisbon

RESEARCHER

Lisbon, Portugal

09/2018 - 12/2023

Teaching and Supervision Experience

School of Business Informatics and Mathematics, University of Mannheim

TEACHING ASSISTANT OF KNOWLEDGE GRAPHS AND DATA MINING

Mannheim, Germany

09/2024 - Present

Faculty of Sciences, University of Lisbon

TUTOR OF MASTER THESIS

Lisbon, Portugal

09/2020 - 09/2024

- Predicting Gene-Disease Associations with Knowledge Graph Embeddings over Multiple Ontologies (Susana Nunes, MSc. in BBC).
- Combining Deep Learning and Knowledge Graphs in PPI Network Prediction Tasks (Laura Balbi, MSc. in BBC).
- Learning with uncertainty: improving supervised learning of protein-protein interactions with lower quality examples (André Mendes, MSc. in Informatics Engineering).

Católica Lisbon School of Business and Economics

TEACHING ASSISTANT OF DATABASE MANAGEMENT

Lisbon, Portugal

02/2022 - 07/2023

Projects

Modelling and mapping of ontologies for cross-sectorial and public health crisis management

European Commission Joint
Research Centre, Italy

EXPERT

06/2025 - Present

- Funded by the European Commission Joint Research Centre
- The goal is to define domain and application ontologies and link them to existing standards, to support semantic interoperability across different EU systems (namely ATHINA, EIOS).

KI-DiabetesDetektion

University of Mannheim, Germany

RESEARCHER

01/2024 - Present

- Funded by Bundesministerium für Bildung und Forschung.
- The goal is to integrate indication data from various sources in a knowledge graph and apply machine learning methods to improve the early-stage detection of diabetes (www.uni-mannheim.de/dws/research/projects/ki-diabetesdetektion).

Serving up-to-date Dynamic Knowledge Graph Embeddings

University of Mannheim, Germany

PRINCIPAL INVESTIGATOR

01/2025 - 06/2025

- Funded by the Open Science Office of the University of Mannheim.
- The goal is to develop a framework capable of periodically collecting new knowledge graph versions, computing embeddings and making them publicly available for further research in a web application (www.uni-mannheim.de/dws/research/projects/serving-up-to-date-dynamic-knowledge-graph-embeddings).

KATY: Knowledge At the Tip of Your fingers

University of Lisbon, Portugal

PHD STUDENT SCHOLARSHIP

11/2023 - 12/2023

- Funded by the European Union's Horizon 2020 research and innovation programme.
- The project aims to build a precise, personalised medicine system empowered by artificial intelligence, which predicts the response of kidney cancer to targeted therapies (<https://katy-project.eu>).

SMiLax: Semantic Mining with Linked Data

University of Lisbon, Portugal

MASTER STUDENT SCHOLARSHIP (PTDC/EEI-ESS/4633/2014)

11/2018 - 08/2019

- Funded by Fundação para a Ciência e a Tecnologia.
- The goal is to improve the state of the art in semantic data mining by establishing novel methods and algorithms for the automated semantic annotation and enrichment of data, whose output will be explored by novel data mining approaches capable of capitalizing on the semantic web (<https://smilax-project.github.io>).

Publications

Relevant Full Papers:

Sousa, Rita T.; Paulheim, Heiko. Improving Knowledge Graph Embeddings through Contrastive Learning with Negative Statements. International Conference on Knowledge Capture (K-CAP), 2025. (Core B)

Sousa, Rita T.; Pesquita, Catia; Paulheim, Heiko. Towards leveraging explicit negative statements in knowledge graph embeddings. Journal of Web Semantics, 2025. (Q2 Scimago)

Sousa, Rita T.; Paulheim, Heiko. Multi-dataset and transfer learning using gene expression knowledge graphs for patient diagnosis. Extended Semantic Web Conference (ESWC), 2025. (Core B)

Sousa, Rita T.; Paulheim, Heiko. Gene Expression Knowledge Graph for Patient Representation and Diabetes Prediction. Journal of Biomedical Semantics, 2025. (Q2 Scimago)

Sousa, Rita T.; Silva, Sara; Pesquita, Catia. Explaining protein-protein interactions with knowledge graph-based semantic similarity. Computers in Biology and Medicine, 2024. (Q1 Scimago)

Nunes, Susana; **Sousa, Rita T.; Pesquita, Catia. Multi-domain Knowledge Graph Embeddings for Gene-Disease Association Prediction.** Journal of Biomedical Semantics, 2023. (Q2 Scimago)

Sousa, Rita T.; Silva, Sara; Paulheim, Heiko; Pesquita, Catia. Biomedical Knowledge Graph Embeddings with Negative Statements. International Semantic Web Conference, 2023. (Core A)

Sousa, Rita T.; Silva, Sara; Pesquita, Catia. Explainable representations for relation prediction in knowledge graphs. International Conference on Principles of Knowledge Representation and Reasoning, 2023. (Core A*)

Sousa, Rita T.; Silva, Sara; Pesquita, Catia. **Supervised Biomedical Semantic Similarity**. IEEE Access, 2023. (Q1 Scimago)

Cardoso, Carlota; **Sousa, Rita T.;** Köhler, Sebastian; Pesquita, Catia. **A collection of benchmark data sets for knowledge graph-based similarity in the biomedical domain**. Database: The journal of biological databases and Curation, 2020. (Q1 Scimago)

Sousa, Rita T.; Silva, Sara; Pesquita, Catia. **Evolving knowledge graph similarity for supervised learning in complex biomedical domains**. BMC Bioinformatics 21 1, 2020. (Q1 Scimago)

Relevant Workshop Papers and Posters:

Ahmad, Hamid; Paulheim, Heiko; **Sousa, Rita T.** **Bio-KGvec2go: Serving up-to-date Dynamic Biomedical Knowledge Graph Embeddings**. International Semantic Web Conference Poster and Demo Track, 2024.

Sousa, Rita T.; Paulheim, Heiko. **Integrating Heterogeneous Gene Expression Data through Knowledge Graphs for Improving Diabetes Prediction**. Semantic Web Solutions for Large-Scale Biomedical Data Analytics Workshop at the Extended Semantic Web Conference, 2024.

Sousa, Rita T.; Silva, Sara; Pesquita, Catia. **Benchmark datasets for biomedical knowledge graphs with negative statements**. Semantic Web Solutions for Large-Scale Biomedical Data Analytics Workshop, 2023.

Sousa, Rita T.; Silva, Sara; Pesquita, Catia. **Towards Supervised Biomedical Semantic Similarity**. Semantic Web Solutions for Large-Scale Biomedical Data Analytics Workshop at the Extended Semantic Web Conference, 2022.

Sousa, Rita T.; Silva, Sara; Pesquita, Catia. **The Supervised Semantic Similarity Toolkit**. Extended Semantic Web Conference Poster and Demo Track, 2022. (Core A)

Nunes, Susana; **Sousa, Rita T.;** Serrano, Filipa; Branco, Ruben; Soares, Diogo F.; Martins, Andreia S.; Auletta, Eleonora; Castanho, Eduardo N.; Madeira, Sara C.; Aidos, Helena; Pesquita, Catia. **Explaining artificial intelligence predictions of disease progression with semantic similarity**. CLEF, 2022.

Sousa, Rita T.; Silva, Sara; Pesquita, Catia. **Is there data leakage in protein-protein interaction prediction using knowledge graphs?**. International Semantic Web Conference Poster and Demo Track, 2021. (Core A)

Sousa, Rita T.; Silva, Sara; Pesquita, Catia. **evoKGsim+: a framework for tailoring Knowledge Graph-based similarity for supervised learning**. Extended Semantic Web Conference Poster and Demo Track, 2021. (Core A)

Cardoso, Carlota; **Sousa, Rita T.;** Köhler, Sebastian; Pesquita, Catia. **A collection of benchmark data sets for knowledge graph-based similarity in the biomedical domain**. Extended Semantic Web Conference Poster and Demo Track, 2020. (Core A)

Honors and Awards

2025	PhD Thesis Award , FCIências.ID (Associação para a Investigação e Desenvolvimento de Ciências)
2024	Best Researcher Award - PhD Student , LASIGE
2024	Two Distinguished Publications ("Explainable representations for relation prediction in knowledge graphs" and "Supervised Biomedical Semantic Similarity"), LASIGE
2023	Best Poster LASIGE Workshop ("Can ChatGPT explain protein interactions better than Knowledge Graphs?"), LASIGE
2021	Best Poster Award ("evoKGsim+: a framework for tailoring Knowledge Graph-based similarity for supervised learning"), Extended Semantic Web Conference
2021	Two Distinguished Publications ("Evolving knowledge graph similarity for supervised learning in complex biomedical domains" and "A collection of benchmark data sets for knowledge graph-based similarity in the biomedical domain"), LASIGE
2020	Best Poster Award ("A collection of benchmark data sets for knowledge graph-based similarity in the biomedical domain"), Extended Semantic Web Conference
2019	Best Poster LASIGE Workshop ("Evolving meaning: Using Genetic Programming to learn similarity perspectives for mining biomedical data"), LASIGE
2018-2019	Merit Scholarship , Direção-Geral do Ensino Superior (DGES)
2017-2018	Maxdata Informatics Award of Excellence , Faculty of Sciences of University of Lisbon

Skills

- **Experience with programming languages** Python (proficient), Java (familiar), R (familiar)
- **Experience with query languages** SQL, SPARQL
- **Experience with machine learning frameworks** scikit-learn, Pytorch, TensorFlow
- **Experience with working with very large biomedical databases and resources** UniProt, STRING, DisGeNET
- **Languages** Portuguese (native), English (full professional proficiency), French (elementary proficiency), German (elementary proficiency)

Event Organization and Participation

- Chair at the International Conference on Knowledge Capture **2025**;
- Organizer of 8th Workshop on SeWeBMeDA co-located with the Extended Semantic Web Conference **2025 & 2026**;
- Participant at Open Science Day - Uni Mannheim **2024**;
- Organizer of the NEGKNOW Challenge at the International Semantic Web Conference **2024**;
- Consultant at DataFest Germany **2024**;
- Participant at Mannheim Center for Data Science Academic Speed Dating **2024**;
- Program Committee of Poster&Demos Track at International Semantic Web Conference **2023**;
- Member of the Genetic and Evolutionary Computation Conference Local Team **2023**;
- Organizer of Extended Semantic Web Conference Student-only Gathering **2023**;
- Program Committee of Poster&Demos Track at Extended Semantic Web Conference **2023**;
- Participant of Ciências's Open Day for Masters and Doctorates **2023**;
- Participant of Girls in ICT **2022 and 2023**;
- Organizer of the 5th LASIGE Workshop **2020**;
- Organizer of Health Sciences Symposiums **2016 and 2017**;
- Organizer of Anatomy Workshops for Health Sciences Students **2015 and 2016**;
- Member of "Núcleo de Ciências da Saúde da Associação de Estudantes da Faculdade de Medicina da Universidade de Lisboa" **2015/2016 and 2016/2017**;

Extracurricular Activity

- **Running** Discoveries Lisbon Half Marathon (2023); São Silvestre Race 10K (2025 & 2024 & 2023 & 2022); Women's Race 5K (2023 & 2022)
- **Volunteering** Volunteer in ReFood (May 2023 - December 2023); Volunteer in Mozambique with Association Internationale des Étudiants en Sciences Économiques et Commerciales (June 2016 - August 2016); Volunteer in Hospital dos Pequenos (2015); Volunteer in Hospital de São Francisco Xavier (2010 - 2011)